

Monopoly

Martin J. Conyon

Bentley University

Required readings

The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 16** — Monopoly.

Read it alongside these slides.

Where we are going

1. **One seller** — what a monopoly is, and the barriers that keep it alone.
2. **The monopolist's revenue** — why selling more means earning less per unit:
 $MR < P$.
3. **Profit maximization** — the quantity, the price, the profit — and the loss.
4. **Monopoly vs. competition** — the trades that die, and what they were worth.
5. **Price discrimination** — one good, many prices.
6. **What can be done?** — regulating a natural monopoly, with numbers.
7. **Extensions** — the Austrian critique of the whole apparatus.

PART 1

One Seller

The opposite pole

Monopoly. A firm that is the sole seller of a product with no close substitutes.

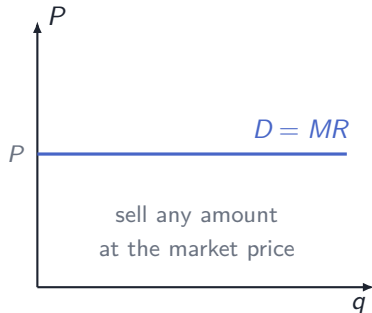
A competitive firm is a **price taker**: the market hands it a price, and its only choice is quantity.

A monopolist is a **price maker**: with no rivals, it chooses the price itself — constrained only by what buyers will take.

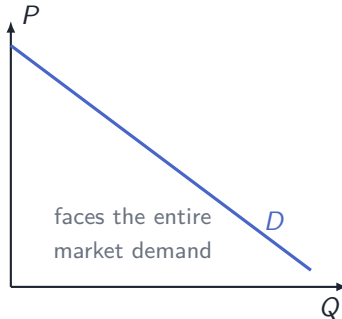
Market power is the ability to raise price without losing every customer.

Two firms, two demand curves

Competitive firm



Monopolist



To sell more, the monopolist must **cut the price** — there is no one else buyers can go to, but there is always the option of *not buying*.

Barriers to entry

Why doesn't profit attract rivals, the way it did in a competitive market? Something blocks the door.

1. **A key resource in one firm's hands** — e.g., control of the ore supply (the aluminum trust broken up in *U.S. v. Alcoa*, 1945). Rare today.
2. **A government grant** — patents, copyrights, licenses: the law itself keeps rivals out.
3. **Natural monopoly** — one firm can serve the whole market more cheaply than two could.

No barrier, no monopoly: free entry always erodes the power.

Barriers by law: the patent trade-off

A patent hands an inventor the exclusive right to sell — typically for 20 years.

- **The cost:** for 20 years, one seller, monopoly pricing — everything measured here.
- **The benefit:** the prospect of those profits is why firms sink fortunes into inventing at all.

Society accepts a *temporary* monopoly as the price of the invention existing.

The monopoly is not a bug in patent law; it is the payment.

Natural monopoly

Natural monopoly. An industry in which a single firm can supply the entire market at lower average cost than two or more firms could.

Our example: **electricity distribution** in a small city — one grid of poles and wires, cost $TC = 160 + 10Q$ (\$ thousands per month, Q = thousands of homes).

$$ATC = \frac{160}{Q} + 10 \quad \text{— falls at every } Q.$$

The fixed cost is the wires; adding one more home costs only \$10. Spreading \$160 over more homes never stops helping.

Why a second grid would be waste

Serve the city's 8,000 homes with $ATC = 160/Q + 10$:

	one firm	two firms
homes per firm	$Q = 8$	$Q = 4$ each
average cost per home	$160/8 + 10 = \text{\$30}$	$160/4 + 10 = \text{\$50}$

A rival would have to **build a second grid** to compete — duplicating the \$160 fixed cost to split the same homes.

Here the barrier to entry is the cost curve itself.

Active Learning 1

For each firm below: is it a monopoly? If yes, **name the barrier** — key resource, government grant, or natural monopoly. If no, **say what substitute or entrant** breaks the power. **The reason is the marks.**

- (a) the only water utility in a city
- (b) a drug maker holding the patent on the only treatment for a disease
- (c) the only steakhouse in a college town
- (d) the owner of the only gravel pit within trucking range of a highway project

Two minutes. Paper and pencil.

PART 2

The Monopolist's Revenue

Write the equation

Our running firm: the **sole maker of a specialized industrial sensor**, protected by a patent. Market demand:

$$P = 100 - 5Q \quad (Q \text{ in thousands per month, } P \text{ in dollars})$$

- Choose $Q = 4$, and the demand curve answers: $P = 100 - 5 \times 4 = 80$.
- Choose $P = 60$, and it answers back: $Q = 8$.

Price maker, yes — but the demand curve is the menu. One choice, not two.

Plug the numbers in

Q	$P = 100 - 5Q$	$TR = P \times Q$	$MR = \Delta TR$
0	100	$100 \times 0 = 0$	—
1	95	$95 \times 1 = 95$	95
2	90	$90 \times 2 = 180$	85
3	85	$85 \times 3 = 255$	75
4	80	$80 \times 4 = 320$	65
5	75	$75 \times 5 = 375$	55
6	70	$70 \times 6 = 420$	45
7	65	$65 \times 7 = 455$	35
8	60	$60 \times 8 = 480$	25
9	55	$55 \times 9 = 495$	15
10	50	$50 \times 10 = 500$	5
11	45	$45 \times 11 = 495$	-5

MR runs below *P* in every row after the first.

Where did the other \$35 go?

The 8th unit sells at $P = 60$. Why did revenue rise by only 25?

To sell 8 rather than 7, the price must fall from 65 to 60 — **on every unit**:

Output effect: one more unit sold, at \$60	+60
Price effect: \$5 less on each of the 7 units already sold	$7 \times (-5) = -35$
net change in revenue ($= MR$, the table's row 8)	+25

To sell one more, cut the price on all — so $MR < P$, always.

Revenue has a summit

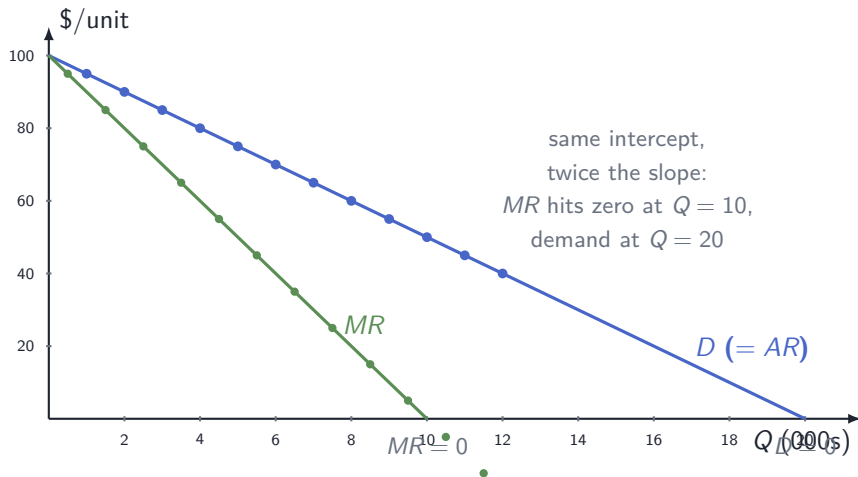
Read the *MR* column again: +5 for the 10th unit, −5 for the 11th.

- Through $Q = 10$: output effect wins — revenue still climbing.
- From $Q = 11$: **price effect wins** — selling more *shrinks* revenue.

Total revenue peaks at $Q = 10$: $TR = 50 \times 10 = 500$.

Beyond the summit, every extra sale costs money. No firm sells there on purpose.

Draw the graph



Two revenue rules

From the table, for *any* firm:

$$AR = \frac{TR}{Q} = \frac{P \times Q}{Q} = P \quad \text{— average revenue is the price.}$$

	competitive firm	monopolist
demand it faces	horizontal at P	the market's, sloping down
price effect	none	always
marginal revenue	$MR = P$	$MR < P$

Next: set this MR against marginal cost — the same output rule every firm obeys.

Active Learning 2

The only licensed shuttle firm at a regional airport faces $P = 80 - 10Q$ (Q = hundreds of rides per day).

- (a) Build the revenue table for $Q = 0$ to 6: columns Q , P , TR , and $MR = \Delta TR$.
Show the multiplication in every row.
- (b) Where does TR peak? What is MR doing there?
- (c) In one sentence: **why** is every MR entry below its price? The why is the marks.

Four minutes. Paper and pencil.

PART 3

Profit Maximization

The cost side of the firm

The sensor firm's total cost, in \$ thousands per month:

$$TC = 200 + 20Q$$

- **Fixed cost** $FC = 200$ — the patent license and the clean room, owed at $Q = 0$.
- **Marginal cost** $MC = 20$ — every extra thousand of sensors adds the same 20.
- **Average total cost** $ATC = 200/Q + 20$, falling as the fixed bill spreads:

Q	2	4	5	8	10	16
ATC	120	70	60	45	40	32.5

Set MR against MC , unit by unit

Each unit is worth making while it adds more than it costs:

unit	MR	MC	$MR - MC$	verdict
1	95	20	+75	make it
2	85	20	+65	make it
⋮ margins shrink \$10 a row ⋮				
7	35	20	+15	make it
8	25	20	+5	make it
9	15	20	-5	skip it
10	5	20	-15	skip it

The firm produces $Q_m = 8$ — where MR meets MC . Same rule as every firm: **produce where $MR = MC$.**

The price comes off the demand curve

$MR = MC$ picks the *quantity*. The *price* is what buyers will pay for it:

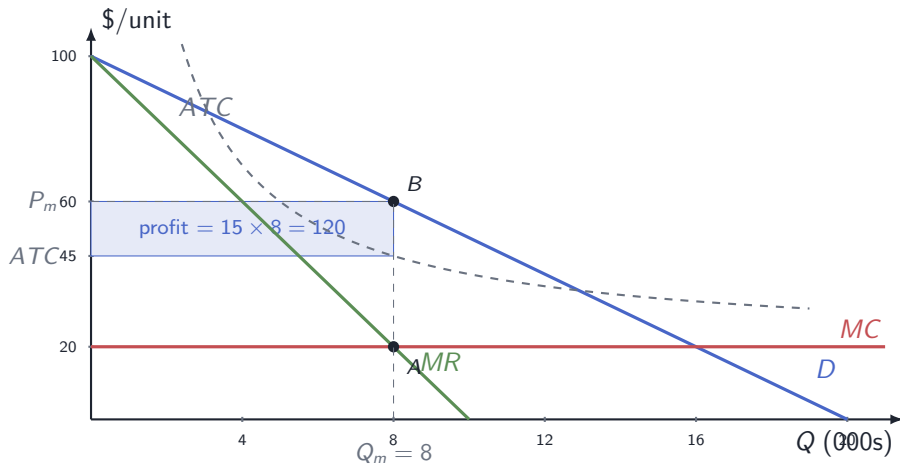
$$P_m = 100 - 5 \times 8 = 60$$

Three steps, always in this order:

1. Find Q_m where $MR = MC$ $\rightarrow Q_m = 8$
2. Go **up to the demand curve** at that quantity $\rightarrow P_m = 60$
3. Never price at the $MR = MC$ crossing — that \$20 is a quantity signal, not a price

$P_m = 60 > MR = MC = 20$ — price above marginal cost.

Draw the graph



The profit, two ways

Route 1 — totals:

$$\pi = TR - TC = 60 \times 8 - (200 + 20 \times 8) = 480 - 360 = \mathbf{120}$$

Route 2 — the rectangle:

$$\pi = (P_m - ATC) \times Q_m = (60 - 45) \times 8 = \mathbf{120}$$

Two routes, one answer — if they disagree, an error is hiding.

\$120,000 a month — and **no entrant is coming to erase it**. The patent holds.
Monopoly profit can persist.

A monopoly can lose money too

Same demand, but suppose the license fee doubles the fixed cost: $TC = 400 + 20Q$.

- $MR = MC$ is untouched by fixed cost $\rightarrow$ still $Q_m = 8$, $P_m = 60$.
- But now $ATC = 400/8 + 20 = 70 > 60$: the rectangle flips **above** the price.

$$\pi = (60 - 70) \times 8 = -\mathbf{80} \quad (\text{check: } 480 - 560 = -80)$$

Market power guarantees a captive market — not a profit.

Why a monopoly has no supply curve

A competitive firm has a supply curve: *for each price, a quantity* — read off MC .

The monopolist never asks “what will I make at a given price?” It asks: **given the whole demand curve, which point do I pick?**

- Change the demand curve, and the same $MC = 20$ yields a different Q_m and P_m .
- Price and quantity are chosen **jointly**, from MR , MC , and demand together.

No fixed price-to-quantity rule exists — so no supply curve.

Active Learning 3

The only ferry licensed on a lake crossing faces

$$P = 40 - 2Q \quad TC = 54 + 4Q \quad (Q = \text{hundreds of crossings/day})$$

- (a) **Predict before computing:** at the profit maximum, will P be above, equal to, or below MC ? Why?
- (b) Find Q_m and P_m . ($MC = 4$ every unit; step the MR column down until it hits it.)
- (c) Compute the profit **both ways** — totals, and the rectangle — and check they agree.

Five minutes. Paper and pencil.

PART 4

Monopoly vs. Competition

What one trade is worth

Rebuild the yardstick in one frame:

- The **demand curve's height** at any Q is what that unit is *worth* to its buyer — the most they would pay.
- **Marginal cost** is what that unit takes to make.
- If value $>$ cost, making it **creates surplus** — value minus cost — split between buyer and seller.

Sensors: unit 4 is worth $100 - 5 \times 4 = 80$ and costs 20 $\rightarrow$ that trade creates **\$60 of surplus**.

Total surplus = value created minus cost, summed over every unit actually made.

The efficient quantity

Make every unit whose value covers its cost — stop where value falls to cost:

$$100 - 5Q = 20 \quad \Rightarrow \quad Q^* = 16$$

A **competitive** sensor industry would land exactly there: price pushed to marginal cost, $P = 20$, $Q = 16$.

Total surplus at $Q = 16$: the triangle under demand above cost,

$$\frac{1}{2} \times (100 - 20) \times 16 = \mathbf{640}.$$

That \$640,000 is the **most this market can deliver** — the benchmark.

The monopolist stops eight units short

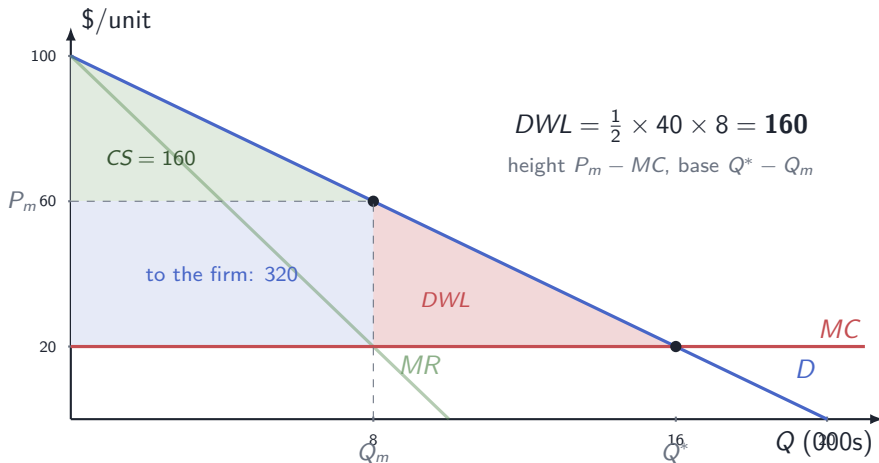
The monopolist chose $Q_m = 8$. Look at the units it **refuses to make**:

unit	worth (demand)	costs (MC)	verdict on the trade
9	55	20	worth making — never happens
12	40	20	worth making — never happens
16	20	20	the last one worth making — never happens

Units 9 through 16: all valued at or above cost, all untraded. **Why?** Selling them would force the price down **on every unit** — the price effect again.

The trades that should have happened, and did not — that is the social cost of monopoly.

Draw the graph — and price the loss



Where the consumers' \$640 went

Under competition, buyers kept the whole \$640. Under monopoly:

kept by consumers	160	surplus on units still bought
transferred to the firm	320	higher price on 8 units — moved, not lost
destroyed	160	the 8 units never made — lost to everyone
640		

The profit is a transfer. The triangle is the loss.

The monopoly price works like a private tax: a wedge between what buyers pay and what production costs — but the “revenue” goes to the firm.

Competition vs. monopoly: the ledger

	competition	monopoly
pricing logic	$P = MC$	$MR = MC$, then $P > MC$
price	\$20	\$60
quantity	16,000	8,000
consumer surplus	640	160
producer surplus	0	320
total surplus	640	480
deadweight loss	0	160
long-run profit	erased by entry	guarded by the barrier

The natural experiment: a patent expires

A brand-name drug's 20-year patent runs out. Generic makers enter the next morning. The market slides from one column of our ledger to the other:

- **Price:** from the monopoly point on demand down toward marginal cost.
- **Quantity:** out from Q_m toward Q^* — patients priced out before are served now.
- **The triangle:** reclaimed — the trades that weren't happening, happen.

Same pill, same factories, same demand — only the barrier changed.

Active Learning 4

The sole licensed operator of airport parking faces

$$P = 60 - 3Q, \quad MC = 12 \text{ (constant)} \quad (Q = \text{hundreds of spaces/day})$$

- (a) **Predict first:** will the monopoly quantity be above or below the efficient quantity? Say why in one sentence.
- (b) Compute both: Q^* from $P = MC$, and Q_m from $MR = MC$ (the MR line falls twice as fast: $60 - 6Q$). Then P_m .
- (c) Compute the deadweight loss. Consumers lose \$288 of surplus in total — **how much of it did the firm get, and how much did no one get?**

Five minutes. Paper and pencil.

PART 5

Price Discrimination

One good, many prices

Price discrimination. Selling the same good to different buyers at different prices, when the cost of serving each buyer is the same.

It takes two ingredients:

1. **Market power** — a price taker cannot charge anyone a premium; rivals undercut it instantly.
2. **No resale** — if bargain buyers can resell to premium buyers, the premium collapses.

Services and licenses discriminate easily (a haircut, a login); goods that ship in boxes do not.

The single price leaves money on the table — twice

At $P_m = 60$, the sensor monopolist walks past two piles of money:

- **The eager buyers.** Unit 2 was worth \$90 to its buyer — who paid 60 and pocketed 30. The firm would love that 30.
- **The excluded buyers.** Units 9–16 are worth 55 down to 20, all above $MC = 20$ — profitable to serve, but only at prices below 60, which the single price forbids.

One price cannot both harvest the top of the demand curve and serve the bottom.

The extreme case: perfect price discrimination

Suppose the firm knew **every buyer's exact valuation** and charged it:

- Every unit worth ≥ 20 gets sold — out to **$Q = 16$** , the **efficient** quantity.
- Revenue = the whole area under demand to 16: $\frac{1}{2}(100 + 20) \times 16 = 960$.
- $\pi = 960 - (200 + 320) = \mathbf{440}$ vs. 120 under one price.

	<i>CS</i>	<i>DWL</i>	total surplus
one price	160	160	480
perfect discr.	0	0	640

Efficient — and the firm keeps every dollar of it.

Real firms approximate it by *group*

No firm reads minds. Real discrimination sorts buyers by an **observable trait** that tracks willingness to pay:

- **Student and senior discounts** — ID cards mark the price-sensitive.
- **Peak vs. off-peak fares** — the schedule-bound pay more; matinee crowds pay less.
- **Coupons and promo codes** — charge less only to those who show they will hunt for it.
- **Versioning** — student vs. professional software editions: same code, two prices.

Each is a coarse screen: two or three prices, not a price per buyer.

Is it good or bad? It depends what it replaces

Compared with the *single monopoly price*:

- **Profit rises** — always. (Otherwise the firm would not bother.)
- **Output can rise** — discount buyers who were excluded get served: the dead triangle shrinks.
- But surplus **shifts from buyers to the firm** — and a clumsy scheme can lower total surplus.

The only guarantee is higher profit. The welfare verdict is case by case.

Active Learning 5

Data-analysis software: $MC = 0$ per copy, $FC = \$4,000/\text{year}$. Buyers:

	how many	each will pay up to
professional firms	100	\$50
students	300	\$10

- (a) One price only: check **both** candidates (\$50 and \$10). Which wins, and what is the profit?
- (b) Now a student edition at \$10 beside the pro edition at \$50: profit?
- (c) Compared with (a): who gains, who loses? **Reason it out — that is the marks.**

Five minutes. Paper and pencil.

PART 6

What Can Be Done?

Four options, one hard case

1. **Antitrust law** — block mergers, break up dominant firms (Sherman Act 1890, Clayton Act 1914). Works where competition *can* exist.
2. **Regulate the price** — the standard tool where it cannot: the natural monopoly.
3. **Public ownership** — run it as a government enterprise.
4. **Do nothing** — if the cure costs more than the disease.

Option 2 sounds easy — “just cap the price at cost.” **Which cost?** The next frames do it with numbers.

The patient: the utility, left alone

From Part 1: $P = 70 - 5Q$, $TC = 160 + 10Q$ (Q = thousands of homes).

$$MC = 10 \quad ATC = \frac{160}{Q} + 10 \quad \Rightarrow \quad \text{MC sits below ATC at every } Q.$$

Unregulated, it behaves like any monopolist (MR falls twice as fast: $70 - 10Q$):

$$70 - 10Q = 10 \Rightarrow Q_m = 6, \quad P_m = 40, \quad \pi = 240 - 220 = 20$$

$$DWL = \frac{1}{2}(40 - 10)(12 - 6) = \mathbf{90} \quad (\text{efficient: } 70 - 5Q = 10 \Rightarrow Q^* = 12)$$

Attempt 1: cap the price at marginal cost

Set $P = MC = 10$. Buyers take $70 - 5Q = 10 \Rightarrow Q = 12$ — the efficient quantity. $DWL = 0$. Perfect? Check the firm's books:

revenue	10×12	120
variable cost	10×12	120
fixed cost (the wires)		160
profit		−160

At $P = MC$, revenue covers the variable cost **exactly** — and not one dollar of the wires. The loss is the entire fixed cost. The firm exits; the regulator has efficiently priced the utility out of existence.

Attempt 2: cap the price at average cost

Let the firm break even: set $P = ATC$ and let the equation pick the point.

$$70 - 5Q = \frac{160}{Q} + 10 \Rightarrow 5Q^2 - 60Q + 160 = 0 \Rightarrow (Q - 4)(Q - 8) = 0$$

Take the larger root (more homes, lower price): **$Q = 8$, $P = 30$** .

Check: $ATC = 160/8 + 10 = 30 = P$ — break even, exactly. The firm survives without a subsidy.

But $P = 30 > MC = 10$: homes 8,000–12,000 still go unserved.

$$DWL = \frac{1}{2}(30 - 10)(12 - 8) = \mathbf{40}$$

The regulator's menu — no free lunch on it

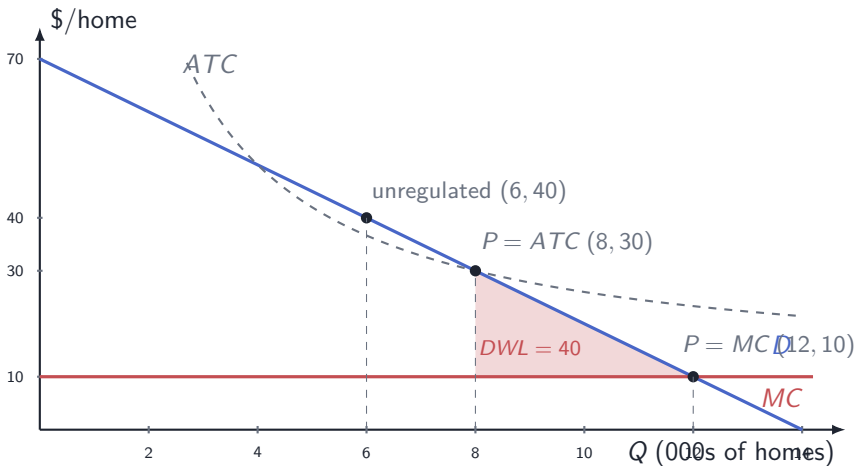
	Q (000s)	P	firm's profit	DWL
unregulated monopoly	6	40	+20	90
cap at ATC (break even)	8	30	0	40
cap at MC (efficient)	12	10	−160	0

Sliding down the menu buys efficiency **with the firm's solvency**:

$P = MC$ is efficient but bankrupts; $P = ATC$ survives but wastes. Pick your poison.

Real rate regulation lives between the last two rows — plus a quieter cost: a firm *guaranteed* break-even has little reason to cut costs.

Draw the graph: the whole dilemma in one picture



Options 3 and 4: own it, or leave it alone

Public ownership. The city runs the grid itself and can price at MC , eating the loss openly through taxes.

- The catch: no owner hunting for cost cuts — losses land on taxpayers, and padding survives.

Do nothing. If DWL is small and regulators are imperfect, tolerating the monopoly can be the least-bad option.

- Stigler's warning: “political failure” from imperfect regulation can cost more than the “market failure” it treats.

Four options, four flaws. Policy picks the cheapest flaw, not a free fix.

Active Learning 6

A city's water utility: $P = 50 - 2Q$, $TC = 150 + 10Q$ (Q = thousands of homes).

- (a) **Predict before computing:** if the regulator caps the price at MC , will the firm profit or lose — and can you name the *size* of the result before doing any arithmetic? Why?
- (b) Now compute it: Q , P , and the profit at $P = MC$.
- (c) Cap at ATC instead: solve $50 - 2Q = 150/Q + 10$ (it factors), take the pro-consumer root, and **verify** $P = ATC$ there. What DWL remains? [Efficient $Q^* = 20$.]

Five minutes. Paper and pencil.

A third case: monopolistic competition

A monopoly has **one** seller and **barriers** that keep it alone. Perfect competition has **many** sellers of an **identical** product.

Between them sits a third case, and it is where a great deal of what you actually buy comes from: **many sellers, each offering a slightly different product.**

- Because the products differ, each firm faces its *own* downward-sloping demand curve. In the short run it behaves exactly like the monopolist we have just studied: it sets $MR = MC$ and reads the price off its demand curve.
- **But entry is free.** Nothing keeps rivals out.

That one difference changes the long run completely. A monopoly keeps its profit because entry is blocked. Here, profit attracts entrants, each takes customers away, and the profit is competed to nothing.

PART 7

Extensions: The Austrian Critique

Competition as a process, not a state

This model is a **snapshot**: one market, one moment, deadweight loss measured against a frozen benchmark.

Hayek (1948) and Kirzner (1973): that confuses competition's *outcome* with competition itself —

- Competition is a **process of discovery**: entrepreneurs hunting profit, innovating, adjusting.
- A market with one seller may still be disciplined — **by the threat of entry**.

Baumol (1982) formalized it: in a **contestable** market (costless entry *and* exit), even a lone incumbent prices near cost. **What matters is the door, not the head-count.**

Schumpeter (1942): the competition that matters is not price rivalry among clones — it is the **new product** that makes the old one obsolete.

- Monopoly profit is the **prize** that funds and motivates innovation — the patent bargain, economy-wide.
- Monopolists are rarely undercut; they are **leapfrogged**.

On this view, today's deadweight triangle buys tomorrow's new industries — tax it away and the engine slows.

The knowledge problem — and the state's own hand

Two more thrusts:

- **The counterfactual is invented.** Our $DWL = 160$ used the competitive price of a market that *does not exist*. Hayek: prices are discovered by competition — absent the process, the “efficient” benchmark cannot simply be looked up.
- **Follow the barrier.** Rothbard (1962): the durable monopolies are mostly **government-made** — patents, licenses, franchises. Private walls crumble; legal walls are maintained. The fix is repeal, not regulation.

What survives the critique

The critique lands real blows: the model is **static**, the benchmark **unobservable**, innovation **absent**.

But it overreaches as a blanket pardon:

- Some monopolies persist for *decades* — network effects, infrastructure, and yes, government protection.
- Creative destruction takes time; consumers pay the markup **while they wait**.

The static model is incomplete, not wrong: withheld output and killed trades are real.
Judge each monopoly with both lenses.

Market power is the power to withhold.

Price above marginal cost kills trades worth making —
and wherever entry is free, entry is what
brings them back.